

Md. Jubayer Hossain

Visiting Scholar, Department of Public Health, Daffodil International University

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Research Statement

I am a computational biologist with a track record of building and deploying reproducible multi-omics frameworks to decode the molecular architecture of cancer and neurological diseases. My research addresses three interconnected questions: (1) How can integrative analysis of bulk RNA-seq, single-cell RNA-seq, and spatial transcriptomics identify robust immunological biomarkers and therapeutic targets within tumor microenvironments? (2) Can interpretable machine learning models trained on pan-cancer transcriptomic data reliably stratify patient prognosis and predict immunotherapy response? (3) What computational strategies best characterize dynamic cell-state transitions—such as tumor-associated macrophage reprogramming and microglial activation in neurodegeneration? My published work includes first- and corresponding-author pan-cancer analyses in *Scientific Reports* and *BMC Genomics*, and seven GitHub-hosted reproducible pipelines built on Nextflow, Docker, and nf-core standards. As a GBD Senior Collaborator at IHME, I have contributed to large-scale epidemiological analyses published in *The Lancet*. In my PhD, I aim to develop scalable, interpretable computational methods for cell-state inference and immune landscape modeling, translating multi-omics discoveries into clinically actionable biomarkers and therapeutic hypotheses.

Education and Training

Master of Science in Microbiology

Dhaka, Bangladesh

Jagannath University

2019–2020

- “CGPA: 3.56/4.0 | Medium of Instruction: English”
- “Capstone Project: Chikungunya Outbreak Management Strategies — Investigated vector surveillance, genomic surveillance, clinical epidemiology, and outbreak response frameworks for arboviral disease in urban Bangladesh”
- “Relevant Coursework: Genomics, Proteomics, Bioinformatics, Research Methodology and Scientific Writing”

Bachelor of Science in Microbiology

Dhaka, Bangladesh

Jagannath University

2016–2019

- “CGPA: 3.15/4.0 | Medium of Instruction: English”
- “Capstone Project: State of the Art of Microbial Genomics Analysis — Reviewed and applied contemporary computational approaches for genomic data analysis including quality control, sequence alignment, variant calling, and visualization of microbial genomic dataset”
- “Relevant Coursework: Biostatistics, Public Health and Hygiene, Bioinformatics I, Bioinformatics II”

Certificate Program in Interdisciplinary Computational Biology

Dhaka, Bangladesh

BRAC University

January 2026

- Mastered protein structure prediction and molecular visualization using AlphaFold2/3 and ChimeraX to investigate protein-ligand interactions
- Applied AI/ML frameworks and docking protocols to structural data from PDB and UniProt for predictive molecular modeling

Specialized Training in AI for Public Health

Dhaka, Bangladesh

Child Health Research Foundation (CHRF)

Sep 2023

- Developed predictive models and data pipelines for disease surveillance, focusing on typhoid incidence and vaccine impact assessment in Bangladesh
- “Acquired Skills: Epidemiological Modeling, Disease Surveillance using AI, Time Series Analysis (LSTM, ARIMA), Medical Image Analysis (X-ray, MRI)”

Technical Skills

- **Programming Languages:** Python, R, Julia, SQL, Bash, JavaScript

- **Statistical Software:** SAS, SPSS, R (Bioconductor, tidyverse), Python (statsmodels, scipy)
- **Bioinformatics:** Bulk RNA-Seq, Single-Cell RNA-Seq, Spatial Transcriptomics, Multi-omics, Microbiome Analysis (16S rRNA)
- **Workflow Management & Reproducibility:** Nextflow (DSL2), nf-core, Docker, Singularity, Git, GitHub
- **Machine Learning & AI:** Scikit-learn, XGBoost, SHAP, PyTorch, Transformers (HuggingFace), DeepChem
- **Molecular Tools:** AlphaFold2/3, ChimeraX, RDKit, BioPython
- **Computing Environments:** UNIX/Linux, HPC clusters, Cloud (AWS, Google Cloud)

Research Experience

Visiting Scholar

Department of Public Health, Daffodil International University

Dhaka, Bangladesh

Sep 2025 – Present

- Investigated computational frameworks within public health genomics to facilitate large-scale biomarker discovery and clinical validation.
- Formulated two research manuscripts (in preparation) analyzing genomic surveillance data and identifying novel molecular signatures in disease.
- Developed a specialized curriculum for AI-driven disease modeling, integrating neural network architectures into biomedical informatics.
- Synthesized heterogeneous digital health records with genomic datasets to evaluate predictive models for clinical outcomes.
- Led a multidisciplinary research initiative to translate raw sequencing data into evidence-based public health insights.

Research Associate (Bioinformatics)

Bio-Bio-1:Bioinformatics Research Discussion Group

Dhaka, Bangladesh

2017 – 2019

- Investigated emerging bioinformatics methodologies, focusing on the application of computational tools to genomic and transcriptomic datasets.
- Designed collaborative training modules in R and Python for data analysis, fostering foundational skills in computational biology for early-career researchers.
- Facilitated technical discussions on sequence alignment algorithms and structural biology, contributing to the group's research output.
- Developed learning content for high-throughput data processing, bridging the gap between theoretical biology and practical data science.
- Evaluated peer-reviewed literature for method validation, maintaining a rigorous standard for the group's computational research initiatives.

Research Intern

International Center for Diarrhoeal Disease Research, Bangladesh

Dhaka, Bangladesh

Feb 2022–Jun 2022

- Designed and implemented a comprehensive search strategy across PubMed, Embase, and PsycINFO, screening 8,000+ articles to support a systematic review on intimate partner violence in South Asia.
- Streamlined the abstract screening process in Covidence, identifying 400+ potentially relevant studies.
- Conducted full-text screening of 120+ articles and extracted key variables for meta-analysis.
- Drafted a systematic review protocol based on PRISMA guidelines, ensuring methodological transparency and reproducibility.

Data Enumerator

School of Allied Health, Curtin University

Australia

Oct 2022–Nov 2022

- Collected both quantitative and qualitative data from 40 Bangladeshi doctors for a Delphi study on Patient Outcomes Measurement in Interprofessional Tuberculosis Care.
- Validated and cleaned all survey responses, achieving 100% accuracy in the final dataset.
- Enhanced the reliability of study outcomes by ensuring high-quality and complete data capture.

Selected Publications

*Note: Bold indicates the co-author; * indicates corresponding author*

1. Ahmed, M. Z., Billah, M. M., Ferdous, J., & **Hossain, M. J.*** (2025). Pan-cancer analysis reveals immunological and prognostic significance of CCT5 in human tumors. *Scientific Reports*, 15, 14405. <https://doi.org/10.1038/s41598-025-88339-z>
2. Fariha, F. T. J., Fuad, M., Saha, C. S., Hossen, S., & **Hossain, M. J.*** (2025). Comprehensive bioinformatics analysis reveals prognostic significance and immunological roles of WNT gene family in breast cancer. *Scientific Reports*, 15, 34490. <https://doi.org/10.1038/s41598-025-13315-6>
3. Bari, S. M., Fuad, M., **Hossain, M. J.** et al. (2025). A meta-analysis of public RNA-Seq data identifies conserved stress responses in rainbow trout. *BMC Genomics*, 26, 999. <https://doi.org/10.1186/s12864-025-12127-2>
4. Shanta, A. S., Islam, N., Al Asad, M., Akter, K., Habib, M. B., **Hossain, M. J.**, Nahar, S., Godman, B., & Islam, S. (2024). Resistance and co-resistance of metallo-beta-lactamase genes in diarrheal and urinary-tract pathogens in Bangladesh. *Microorganisms*, 12(8), 1589. <https://doi.org/10.3390/microorganisms12081589>

Papers Under Review

1. Islam, M. W., Rahman, M. M., Naznin, H., Hossain, M. S., Akter, T., Shatabde, Z. A., & **Hossain, M. J.*** (2025). “Integrative bioinformatics analysis reveals COL13A1 and COL23A1 as potential diagnostic and prognostic biomarkers in thyroid cancer”. *Under Review*.
2. Islam, M. W., Fariha, F. T. J., Ahmed, M. Z., Ferdous, J., Lota, H. B., Hossain, M. S., Kundu, P., Shahjahan, M., & **Hossain, M. J.*** (2025). “Bioinformatics-driven multi-omics profiling of CDK1 and CDK6 identifies prognostic and therapeutic roles in breast cancer”. *Under Review*.
3. Mabsurah, K., Yeana, I. J., Sumaiya, M., Islam, T., Ahammad, K., Bhattacharjee, A., Ferdous, J., Billah, M. M., & **Hossain, M. J.*** (2025). “Pan-cancer analysis identifies potential prognostic and immunological role of RAS-degrading WDR76 in human tumors”. *Under Review*.
4. Kundu, P., Shahariar, M., Munny, F. A., Riya, A. T., Lota, H. B., Ahmed, M. Z., Meme, M. A., Kona, U. M. P., Mahmuda, F., Chowdhury, S. S., Sraboni, S. A., & **Hossain, M. J.** (2026). Computational identification of *Taxus baccata*-derived phytochemicals targeting EZH2 to overcome therapeutic resistance in melanoma. *Research Square*. <https://doi.org/10.21203/rs.3.rs-8494315/v1>. *Under Review*.

Research in Progress

1. **Hossain, M. J.**, Shahjahan, M., & Fuad, M. (2025). Fezf2-mediated cortical development: Multi-omics analysis. *Working Paper/GitHub*. <https://github.com/hossainlab/fezf2-multiomics>
2. **Hossain, M. J.**, Shahjahan, M., & Fuad, M. (2025). Single-cell meta-analysis of microglial activation in Alzheimer’s disease. *Working Paper/GitHub*. <https://github.com/hossainlab/alzheimers-microglia-meta-analysis>
3. **Hossain, M. J.**, Shahjahan, M., & Fuad, M. (2025). Integrative spatial-scRNA-seq atlas of immunotherapy resistance mechanisms across cancer types. *Working Paper/GitHub*. <https://github.com/hossainlab/spatial-scRNA-immune-resistance>
4. **Hossain, M. J.**, Shahjahan, M., & Fuad, M. (2026). Pan-Cancer Tumor-Associated Macrophage Reprogramming Atlas and Immunotherapy Response Prediction. *Working Paper/GitHub*. <https://github.com/hossainlab/TAM-Atlas>
5. DeepAMR: AI-powered antimicrobial resistance prediction from genomic data. *Working Paper/GitHub*. <https://github.com/hossainlab/DeepAMR>

Selected Conferences

1. Ahmad, F., Fuad, M., Shahjahan, M., & **Hossain, M. J.** (2025). Identification of Subtype-Specific Prognostic Biomarkers in Esophageal Carcinoma through Integrated Bioinformatics Analysis. 9th BSM Annual Conference, University of Dhaka, Bangladesh. *GitHub*. https://github.com/gsabioinfornternship/ESCA_Meta

- Aditee, L. M., Fuad, M., Shahjahan, M., & **Hossain, M. J.** (2025). Network-Driven Bioinformatics Approach Uncovers Prognostic Signatures and Therapeutic Targets in Pancreatic Ductal Adenocarcinoma. 9th BSM Annual Conference, University of Dhaka, Bangladesh. *GitHub*. https://github.com/gsabioinfointernship/PAAD_Meta
- Ahamed, M. S., Fuad, M., Shahjahan, M., & **Hossain, M. J.** (2025). Bioinformatic Analysis of Transcriptomic Data Reveals Molecular Signatures Driving Colorectal Cancer. 9th BSM Annual Conference, University of Dhaka, Bangladesh. *GitHub*. https://github.com/gsabioinfointernship/COADREAD_Meta
- Tabassum, R., Fuad, M., Shahjahan, M., & **Hossain, M. J.** (2025). Comparative Transcriptomic Meta-Analysis Reveals Shared and Distinct Transcriptional Networks and Pathway Crosstalk in Hepatocellular Carcinoma for Biomarker and Therapeutic Target Identification. 9th BSM Annual Conference, University of Dhaka, Bangladesh. *GitHub*. https://github.com/gsabioinfointernship/LIHC_Meta

Professional Recognition

- Nextflow Ambassador** | Seqera, Barcelona, Spain | 2025–Present Selected from the global bioinformatics community to promote reproducible, scalable workflow development using Nextflow DSL2.
- GBD Senior Collaborator** | Institute for Health Metrics and Evaluation (IHME), University of Washington | 2025–Present Contributing analyst to three *Lancet*-family GBD 2023 publications on vaccine coverage, headache burden, and breast cancer burden.
- Expert Media Commentary** | The Business Standard, Bangladesh | 2024 Cited as subject-matter expert in two national health features on antibiotic resistance and self-medication practices.

Teaching Experience

Lead Organizer & Instructor

Dhaka, Bangladesh

DeepBio Academy, Bioinformatics and Computational Biology

June 2020 – Present

- Designed comprehensive bioinformatics curriculum specializing in Single-Cell RNA-seq (scRNA-seq) workflows and multi-omic data integration.
- Benchmarked computational tools for transcriptomics and genomics to ensure pipeline reproducibility and analytical rigor.
- Supervised a technical research team in processing high-throughput omics datasets to identify transcriptional drivers of disease.
- Mentored undergraduate students in the development of computational research portfolios, emphasizing open-science standards and version control.
- Refined deep learning modules for biological applications, focusing on the interpretability of high-dimensional data reduction techniques.

Faculty (Bioinformatics)

Dhaka, Bangladesh

University of Dhaka (cBLAST)

Aug 2023 – Present

- Instructed specialized modules in Biomedical Machine Learning, focusing on the application of Python-based neural networks to sequence analysis.
- Directed student research projects applying deep learning frameworks to identify regulatory elements in complex biological datasets.
- Facilitated technical practicums for 100+ trainees on scalable data science methodologies for high-throughput genomic data.
- Evaluated computational frameworks for performance, establishing rigorous criteria for the optimization of biological models.
- Bridge theoretical computational principles with experimental biology to foster collaborative research outputs.

Program Lead

Dhaka, Bangladesh

GSA Bioinformatics Internship

July 2025 – Dec 2025

- Conducted a pan-cancer transcriptomics meta-analysis across four organ systems to identify conserved regulatory networks in tumorigenesis.
- Optimized high-throughput pipelines for Bulk and Single-Cell RNA-seq, implementing advanced batch-effect correction and normalization protocols.

- Mentored research interns in the utilization of High-Performance Computing (HPC) environments and R/Bioconductor analytical frameworks.
- Analyzed complex transcriptional networks using Gene Set Enrichment Analysis (GSEA) to uncover novel oncogenic drivers.
- Managed collaborative research workflows across three institutions, resulting in the generation of peer-reviewed conference abstracts.

Teaching Assistant

Department of Microbiology, Jagannath University

Dhaka, Bangladesh

Oct 2022 – Dec 2022

- Assisted in the delivery of research methodology training, focusing on experimental design and statistical power in public health microbiology.
- Processed public health surveillance data to support departmental research on emerging epidemiological trends.
- Critiqued and refined student research proposals to ensure compliance with modern computational and microbiological standards.
- Communicated complex biological theories through technical presentations to diverse academic audiences.
- Implemented digital data-tracking systems to monitor research milestones and student performance metrics.

Leadership Experience

Founder & Principal Investigator

DeepBio Limited (Bioinformatics Research Group)

Dhaka, Bangladesh

Jan 2025 – Present

- Lead a research team of 13 scientists executing scRNA-seq and bulk RNA-seq analysis pipelines for cancer genomics and transcriptomics.
- Developed and benchmarked scalable bioinformatics workflows for biomarker discovery in cancer and neurodegenerative diseases.
- Built AI-integrated computational infrastructure for multi-omics data processing and interpretation.
- Designed and delivered a national training program in computational biology, upskilling undergraduate researchers across Bangladesh.
- Transitioning from operational leadership to a scientific advisory role to pursue full-time doctoral studies in 2026.

Founder & Capacity Building Director

CHIRAL Bangladesh (Non-profit Research Institute)

Dhaka, Bangladesh

June 2020 – Present

- Founded and lead a research institute with dedicated divisions in Bioinformatics, Public Health, Geospatial Health, and AI-Healthcare.
- Published 20+ peer-reviewed articles in Q1/Q2 journals and 40+ conference papers in health informatics and computational biology.
- Supervised 15+ manuscripts through analytical development and peer-review preparation in global health and bioinformatics.
- Collaborated with Michigan State University on biomedical research capacity-building initiatives in South Asia.
- Trained 3,000+ early-career researchers in data science, research methodology, and computational biology workflows.

Community Engagement

Nextflow Ambassador

Seqera

Barcelona, Spain

2025 – Present

- Advocated for the adoption of Nextflow-based DSL2 pipelines to ensure reproducibility and scalability in high-throughput single-cell and bulk RNA-seq workflows.
- Engineered technical tutorials and documentation for the deployment of portable scientific containers (Docker/Singularity) in cloud and HPC environments.
- Optimized community-driven bioinformatic workflows, focusing on the integration of standardized pipelines for large-scale transcriptomic data processing.
- Facilitated technical workshops on workflow orchestration, training researchers to manage complex multi-omic datasets with high analytical rigor.
- Collaborated with global health data scientists to implement best practices for version-controlled, open-source computational research.

GBD Senior Collaborator

Institute for Health Metrics and Evaluation

Washington, USA

2025 – Present

- Contributed advanced statistical modeling and analytical validation to Global Burden of Disease (GBD) studies published in high-impact journals like The Lancet.
- Synthesized and audited complex epidemiological datasets from 190+ countries to strengthen the robustness of global health genomic evidence.
- Formulated predictive models for disease burden, utilizing large-scale data integration techniques to identify health trends across diverse populations.
- Evaluated the quality and consistency of multi-national health data, ensuring high-fidelity inputs for global health policy frameworks.
- Collaborated with international research cohorts to refine statistical methodologies for assessing neurological and metabolic disease prevalence.

References

Dr. Syeda Tasneem Towhid Associate Professor, Department of Microbiology Jagannath University, Dhaka, Bangladesh Email: towhidst@mib.jnu.ac.bd *Relationship: M.Sc. Project Supervisor and long-term research collaborator (2019–present)*

Dr. Md. Salekul Islam Professor, Department of Microbiology Jahangirnagar University, Dhaka, Bangladesh Email: salekul@juniv.edu *Relationship: Collaborative researcher in antimicrobial resistance and microbiology studies*

Additional references from international collaborators at the Institute for Health Metrics and Evaluation (IHME, University of Washington) and Michigan State University are available upon request.